

Dynamic and static UWB/IMU validation

FlexTDOA, Native DS-TWR and Passive DS-TWR

Field campaign and controlled replay ablation – 11 August 2026

Platform: 5 ESP32-S3 + DW3000 modules; Raspberry Pi collector
Topology: M1 tag; M2, M3, M4 and M5 anchors
Radio: UWB channel 9; configured 40 MHz SPI
Firmware: deployed `bb23b07-dirty`; analysis branch `imu-fusion-smoothing`
Sensor path: BNO085 acquisition near 500 Hz; GPTimer timestamps
Evidence: 3 dynamic + 3 static protocol captures; replay and ablation artifacts

Main result. Raw UWB remains the authoritative positioning stream. The current fusion output can suppress isolated extremes, but controlled replays show that the measurable improvement comes from the common position smoother, not from accelerometer integration. The accelerometer contribution is therefore **not accepted** for production accuracy claims.

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1 Executive summary

This report is the complete counterpart of the 6 August protocol report. It uses the same six-cell matrix (three protocols in dynamic and static modes), the same independent-frame policy, the same local static precision metrics, and the same RTK registration audit. It additionally contains the controlled IMU ablation that is necessary to distinguish inertial information from ordinary trajectory smoothing.

Evidence boundary. Counts, rates, positive gaps and local static clouds are direct diagnostics. Dynamic UWB-RTK errors are useful in-system comparisons, but the UWB packet does not yet carry the radio measurement timestamp. Static absolute ranking is invalid for the original three 30 s captures because their RTK centers differ by 2.03 m and two anchor registrations are poor. No outlier was removed.

The principal results are:

- Static raw local RMS is **2.13 cm** for FlexTDOA, **2.02 cm** for Native DS-TWR and **1.78 cm** for Passive DS-TWR. These values measure repeatability around each capture median.
- Dynamic independent rates are **12.90**, **20.86** and **13.33 positions/s**, respectively. Flex and Passive contain multi-second maximum gaps; Native contains a 1.02 s maximum gap.
- On the controlled Native replay, raw RMSE is 17.911 cm. Disabling acceleration gives 16.323 cm; the current acceleration path gives 16.326 cm. The difference is **+0.0031 cm**: acceleration provides no measurable improvement.
- Passive gives an even stronger null result: disabled, current and relaxed acceleration configurations all produce 14.867 cm RMSE and 29.389 cm P95 to the displayed precision.
- A new Passive stationary control with the full 500 Hz telemetry path yields **2.710 cm raw RMSE versus RTK** and **1.711 cm local raw RMS**. The radio/solver path has therefore not suffered a general stationary regression from enabling high-rate IMU acquisition.

2 Questions and acceptance criteria

The campaign evaluates four separate questions:

1. Does each protocol continue to deliver raw positions at a useful rate?
2. What is the unfiltered stationary repeatability of each protocol?
3. Does the current fused output improve dynamic RMSE and P95 without increasing overshoot or hiding raw data?
4. Is any observed improvement caused by acceleration, rather than by the position smoother shared by all replay variants?

The causal acceptance criterion is intentionally strict: acceleration must beat an otherwise identical replay with acceleration disabled on both RMSE and P95, without a worse maximum or new dead zones. A raw-versus-fused comparison alone cannot prove an IMU gain because the fused branch also includes endpoint smoothing, constant-velocity propagation, reacquisition logic and ZUPT.

3 System and protocol background

3.1 FlexTDOA

The tag is UWB receive-only. Anchors exchange scheduled transmissions and the tag solves from a coherent set of time-difference observations. Airtime scales well with passive tag count. The current capture includes both independent frames and repeated/publication events; only independent frames enter fair rate and trajectory metrics.

3.2 Native DS-TWR

The active tag ranges to every anchor and solves one coherent four-range frame. It offers the clearest measurement semantics but consumes tag airtime. A frame can be incomplete when one or more anchor exchanges miss their deadline.

3.3 Passive DS-TWR

The tag listens to anchor-to-anchor DS-TWR exchanges. Rolling windows can publish multiple positions from shared measurements. Therefore all-event rate and independent-frame rate are reported separately.

3.4 IMU path under test

BNO085 samples carry 64-bit GPTimer time into replay. Orientation association, reordering and interpolation are available, and raw UWB remains permanently separate. The filter can use constant-velocity propagation, position endpoint smoothing, stationary bias estimation, ZUPT and yaw-conditioned horizontal acceleration. That mixture is the reason for the controlled ablation.

4 Captured evidence

The selected dynamic walks lasted 177–243 s; the static blocks lasted about 30 s. The Passive dynamic JSONL lacks its final `capture_end` record, but its 177 s data span and replay are readable. It is included and explicitly marked incomplete rather than silently discarded.

Protocol	Mode	capture ID	duration s	events	independent	indep Hz	coverage %	container
FlexTDOA	dynamic	flextdoa_dynamic_final.01	240.5	4023	3101	12.90	99.4	complete
Native DS-TWR	dynamic	native_ds_dynamic_quality_gate.02	243.0	5069	5069	20.86	99.7	complete
Passive DS-TWR	dynamic	passive_ds_dynamic_final.01	176.9	7033	2358	13.33	100.0	missing capture_end
FlexTDOA	static	flextdoa_static_final	30.3	929	929	30.67	99.4	complete
Native DS-TWR	static	native_ds_static_final	30.5	686	686	22.47	99.4	complete
Passive DS-TWR	static	passive_ds_static_final	31.1	2295	772	24.80	99.1	complete

5 Methodology

5.1 Population and continuity

Interleaved records marked as fused are excluded from raw UWB metrics. Passive and Flex publications are reduced to independent frames where the protocol marks them. Rates use the full capture wall duration. Positive gaps are differences between unique ESP32 uptime values; P50, P95, P99, maximum and counts above 100 ms are retained.

5.2 Static precision

Raw radial displacement is calculated about the component-wise median of each capture. CEP50, RMS, P95, P99, maximum, 2DRMS and first-to-last-decile drift retain every independent sample. These metrics describe local repeatability, not absolute accuracy.

5.3 RTK association and frame registration

Only quality-4 RTK Fixed fixes enter comparisons. WGS84 is converted to ENU. A proper 2-D rigid rotation and translation (no scale) is fitted from anchor geometry, never from the tag trajectory. UWB and RTK samples are associated in software within 100 ms.

The measurement-time estimate currently subtracts reported GNSS fix age from collector receive time. A separate audit found this estimate about 108–112 ms late relative to GNSS UTC; correcting it explains roughly 5–6 cm of dynamic RMSE for Flex and Native, but not the complete 15–18 cm error. The radio measurement timestamp must be transmitted end-to-end before sub-decimetre dynamic claims are treated as final.

5.4 Controlled IMU ablation

Raw, acceleration-disabled, current-gate and relaxed-gate configurations are replayed on the same captured population. The disabled replay preserves the same position smoother and constant-velocity machinery. Consequently, *current minus disabled* isolates the acceleration contribution much more cleanly than *fused minus raw*.

6 Dynamic raw results

6.1 Trajectories

The walks are separate physical trials rather than synchronized repetitions. The panels are therefore used for continuity, jumps and gross geometry, not point-by-point route ranking.

6.2 Update rate and gaps

Protocol	event Hz	indep Hz	P50 ms	P95 ms	P99 ms	max ms	>100 ms	step P95 m/s
FlexTDOA	16.73	12.90	30	160	280	2550	380	14.75
Native DS-TWR	20.86	20.86	40	70	130	1020	71	5.22
Passive DS-TWR	39.75	13.33	10	120	240	2680	385	4.29

Native has the strongest dynamic cadence in this campaign. Flex and Passive both contain multi-second interruptions. The very large Flex step-speed P95 is a direct symptom of abrupt raw position variation, not measured walking speed.

6.3 Native radio-loss diagnostic

Across the Native capture, start/end counter deltas give 4254 complete and 1208 incomplete candidate frames: **22.12% incomplete**. There were 658 slot overruns. All 485 PHY errors were reported as recovered, but the same interval also contained 964 response timeouts, 517 result timeouts, 377 frame sync losses, 90 PHR errors and 18 FCS errors.

Anchor	attempts	completed ranges	completion %	response timeout	result timeout
A2	5462	5028	92.06	302	131
A3	5461	5042	92.33	289	130
A4	5461	5219	95.57	120	122
A5	5461	5074	92.91	253	134

Loss is not confined to one anchor, although A4 is visibly better. The data do not support blaming geometry alone. Scheduling and RX recovery remain credible contributors.

7 Static raw results

7.1 Local precision

Protocol	N	CEP50 cm	RMS cm	P95 cm	P99 cm	max cm	2DRMS cm	drift cm
FlexTDOA	929	1.68	2.13	3.85	4.89	5.95	4.27	0.21
Native DS-TWR	686	1.40	2.02	3.76	5.99	10.06	4.02	0.16
Passive DS-TWR	772	1.50	1.78	3.11	3.72	6.44	3.57	0.45

All three protocols remain close to the requested 2 cm local RMS target at the tested central point. This does not establish 2 cm absolute accuracy across the polygon or under NLOS.

7.2 Static RTK consistency gate

The maximum separation between the three stationary RTK centers is 2.032 m, versus a 0.250 m consistency gate. Flex has no usable anchor transform in its static block. Native and Passive anchor-fit RMS values are 1.073 m and 0.741 m, respectively. Therefore the approximately 3.0 m Native and 1.9 m Passive absolute offsets produced by these blocks are registration artifacts and **must not be used to rank protocols**.

7.3 Current stationary regression control

A new 30 s Passive capture made after the high-rate IMU changes has valid RTK registration and provides the relevant regression check.

Metric	result
Independent raw positions	832
Raw UWB RMSE versus RTK	2.710 cm
Raw UWB P95 versus RTK	4.080 cm
Raw UWB maximum versus RTK	4.993 cm
Raw local-cloud RMS	1.711 cm
Raw local-cloud P95	2.915 cm
Fused RMSE versus RTK	2.568 cm
Anchor rigid-fit RMS	0.422 cm
RTK rate	8.08 Hz
Tag IMU rate used by replay	458.9 Hz

This is close to the prior calibrated stationary baseline of 2.55 cm absolute RMSE. High-rate sensor acquisition and telemetry have not generally degraded stationary raw UWB accuracy.

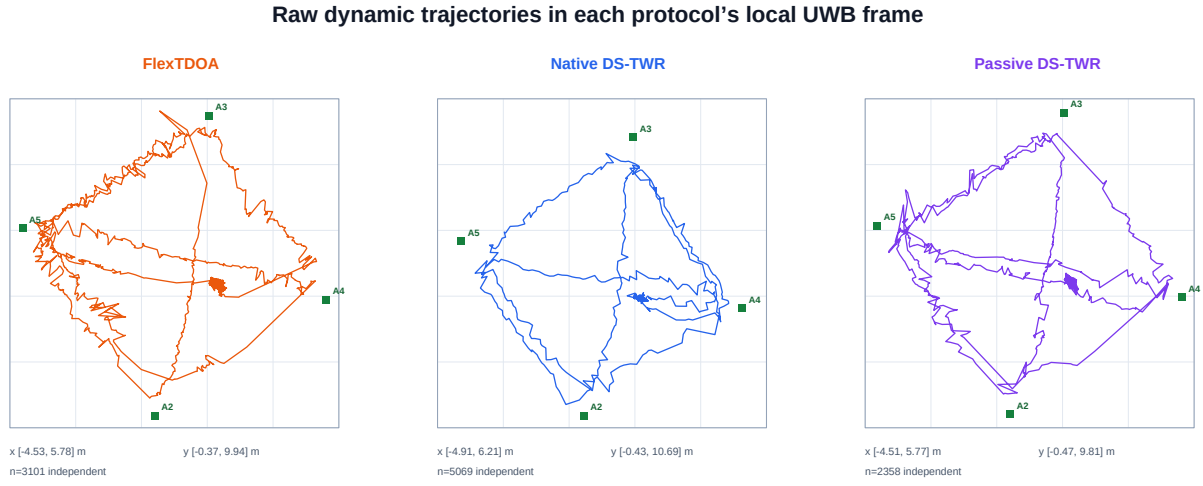


Figure 1: Raw independent dynamic trajectories in each local UWB frame. Green markers are anchors. Outliers are retained.

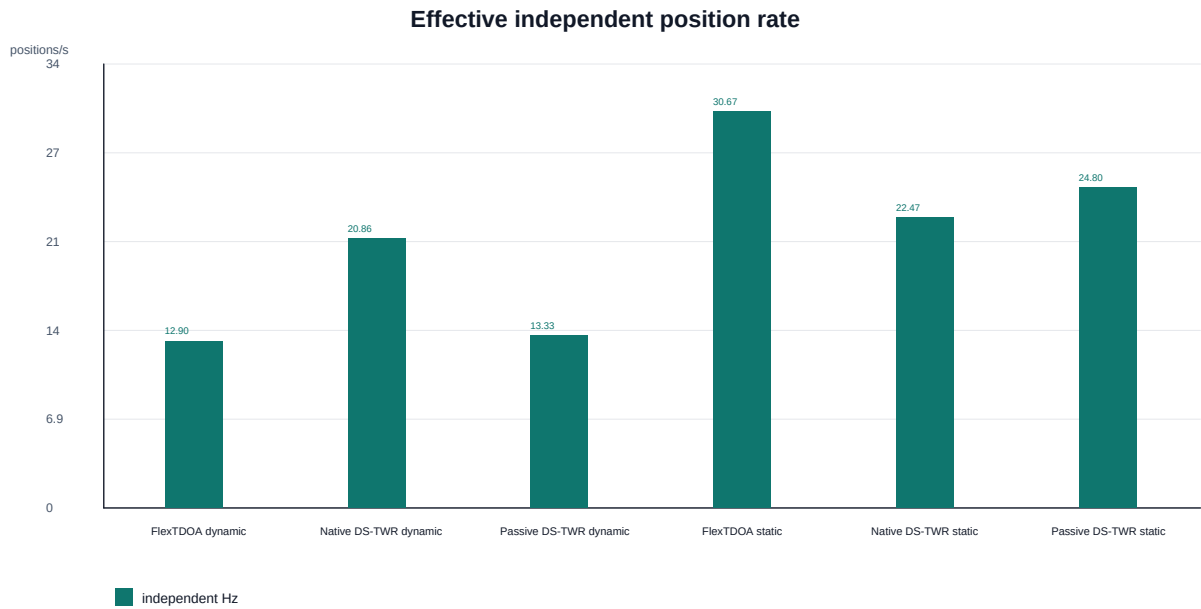


Figure 2: Independent position rate. Passive all-event publication rate is not substituted for independent frames.

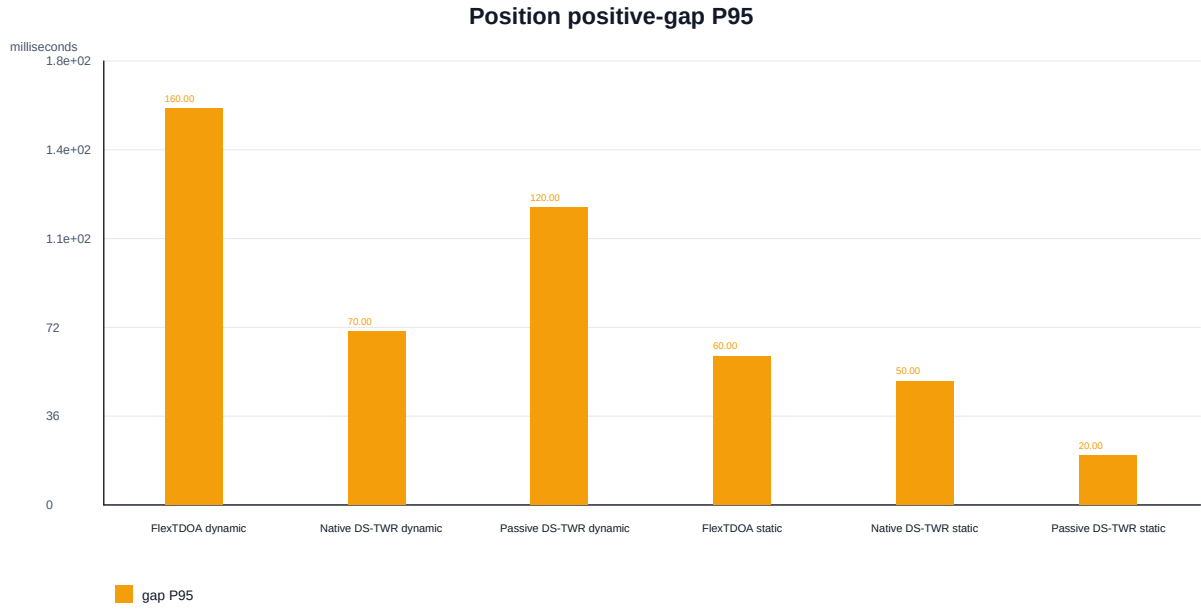


Figure 3: Positive-gap P95 for dynamic and static raw streams. Maximum gaps are listed in the tables.

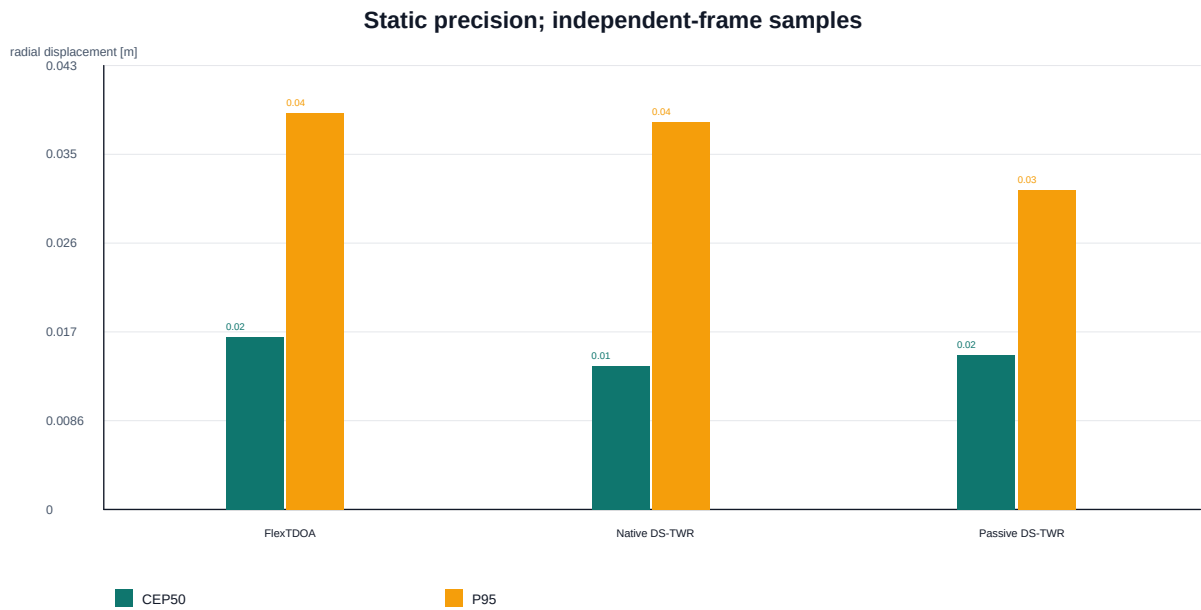


Figure 4: Static raw precision around each capture median. All independent samples are included.

Static raw position clouds about each capture median

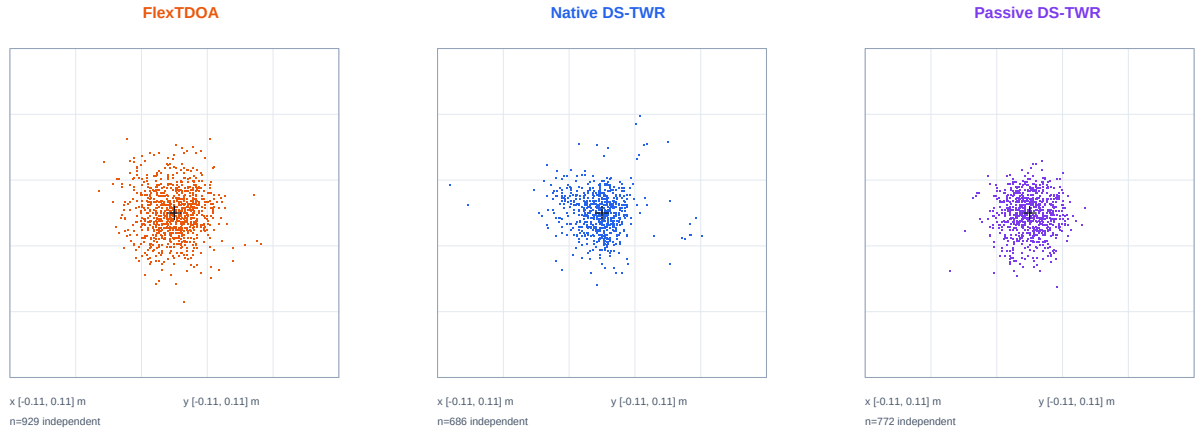


Figure 5: Static raw position clouds centered on each capture median and drawn at one common scale. The cross is the median.

GPS RTK Fixed tag clouds and cross-capture centers

Top: each block centered on its own median. Bottom: the same fixes in one common ENU frame.



Figure 6: Static GPS RTK Fixed tag solutions. Top: each cloud centered on its own median at a shared centimetre scale. Bottom: the same fixes in a common ENU frame, exposing the cross-capture reference displacement.

8 Dynamic UWB–RTK audit

Protocol	pairs	pair coverage %	time P95 ms	raw RMSE cm	raw P95 cm	raw max cm	anchor fit P95 cm	debiased RMSE cm
FlexTDOA	1100	62.6	65.6	18.28	30.57	80.19	1.41	17.16
Native DS-TWR	1892	98.2	28.0	18.68	28.36	278.00	2.25	14.77
Passive DS-TWR	1154	83.3	83.4	16.75	34.66	67.91	1.50	16.82

These values use direct capture-time association and anchor-based rigid registration. Replay tables later use their own identical-timestamp population; small differences in sample selection are expected and are retained rather than forced to agree.

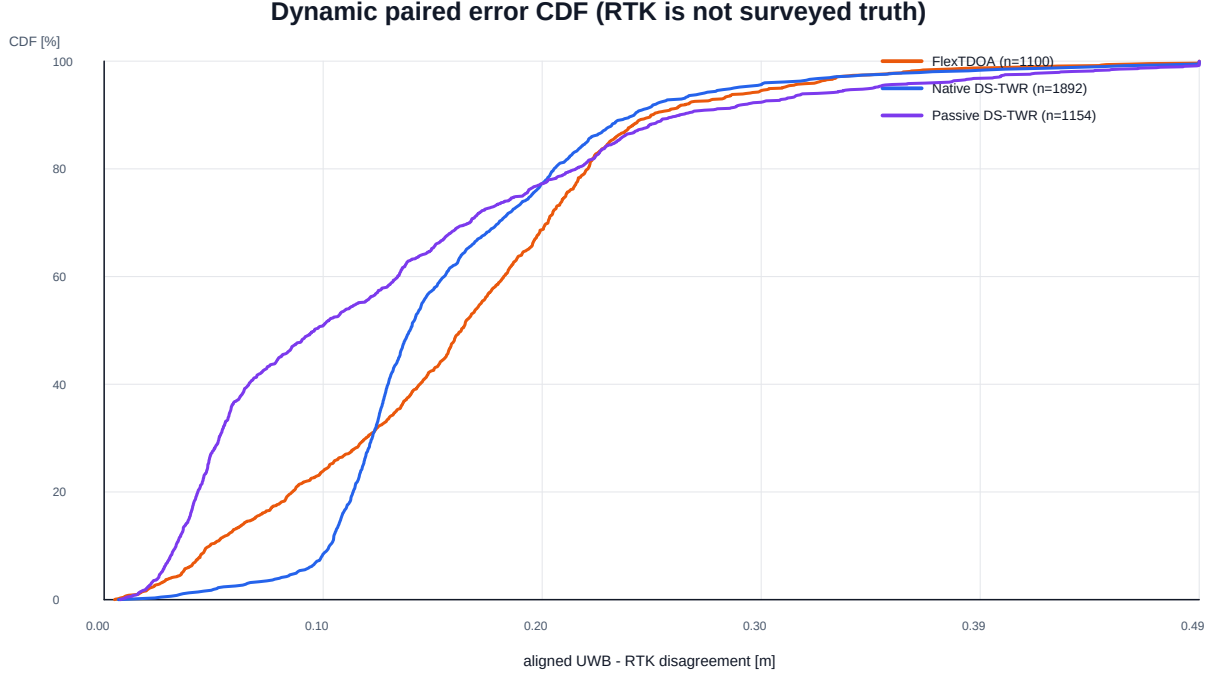


Figure 7: Dynamic raw UWB–RTK disagreement CDF after anchor registration. RTK is an in-system reference, not surveyed truth.

9 Fusion replay and causal ablation

9.1 Raw versus selected fused replay

The table below uses the per-tag comparison population and the selected replay for each protocol. It describes the complete fused algorithm, not acceleration alone.

Protocol	pairs	raw RMSE cm	fused RMSE cm	raw P95 cm	fused P95 cm	raw max cm	fused max cm	gap count delta	gate
FlexTDOA	1044	17.262	16.883	28.062	28.263	78.42	65.12	-190	reject
Native DS-TWR	1860	17.911	16.326	26.822	25.395	260.87	103.49	-57	nominal pass
Passive DS-TWR	1067	14.953	15.029	29.440	29.818	67.78	67.41	-355	reject

Flex reduces RMSE and maximum but increases P95 by 2.0 mm. Passive slightly worsens both RMSE and P95. Native appears to pass a raw-versus-fused gate, but that result is not causal evidence for the accelerometer.

9.2 Controlled acceleration ablation

Protocol	configuration	accel used / valid	RMSE cm	P95 cm	max cm	generic replay pass
Native DS-TWR	raw UWB	–	17.910850	26.821895	260.873390	–
Native DS-TWR	acceleration disabled	4 / 108563	16.323327	25.395069	105.326516	yes
Native DS-TWR	current gates	20449 / 108563	16.326430	25.395069	103.493998	yes
Native DS-TWR	relaxed gates	97763 / 108563	16.310498	25.649909	94.720490	yes
Passive DS-TWR	raw UWB	–	14.952546	29.439575	67.783143	–
Passive DS-TWR	acceleration disabled	0 / 78893	14.867235	29.388807	66.904426	yes
Passive DS-TWR	current gates	0 / 78893	14.867235	29.388807	66.904426	yes
Passive DS-TWR	relaxed gates	26903 / 78893	14.867235	29.388807	66.904426	yes

For Native, current minus disabled is +0.003103 cm RMSE and exactly 0.000 cm P95 to the shown precision. Relaxed acceleration changes RMSE by only -0.012829 cm while worsening P95 by 0.254840 cm. Passive is

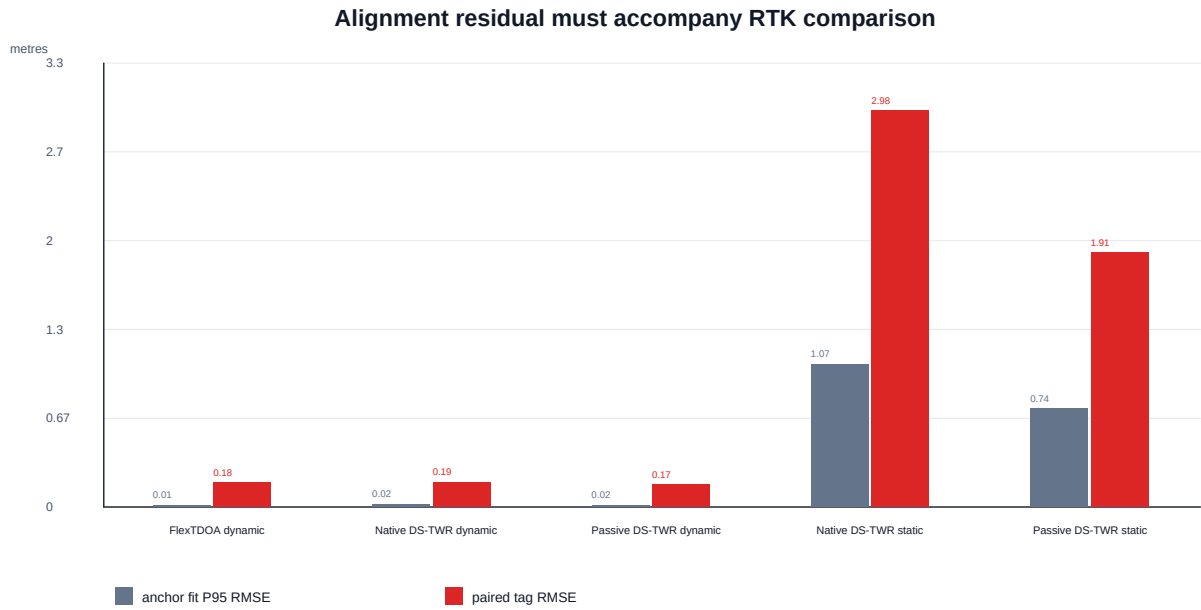


Figure 8: Anchor-registration residual beside paired tag disagreement. Dynamic anchor fits are centimetric; timing remains the larger limitation.

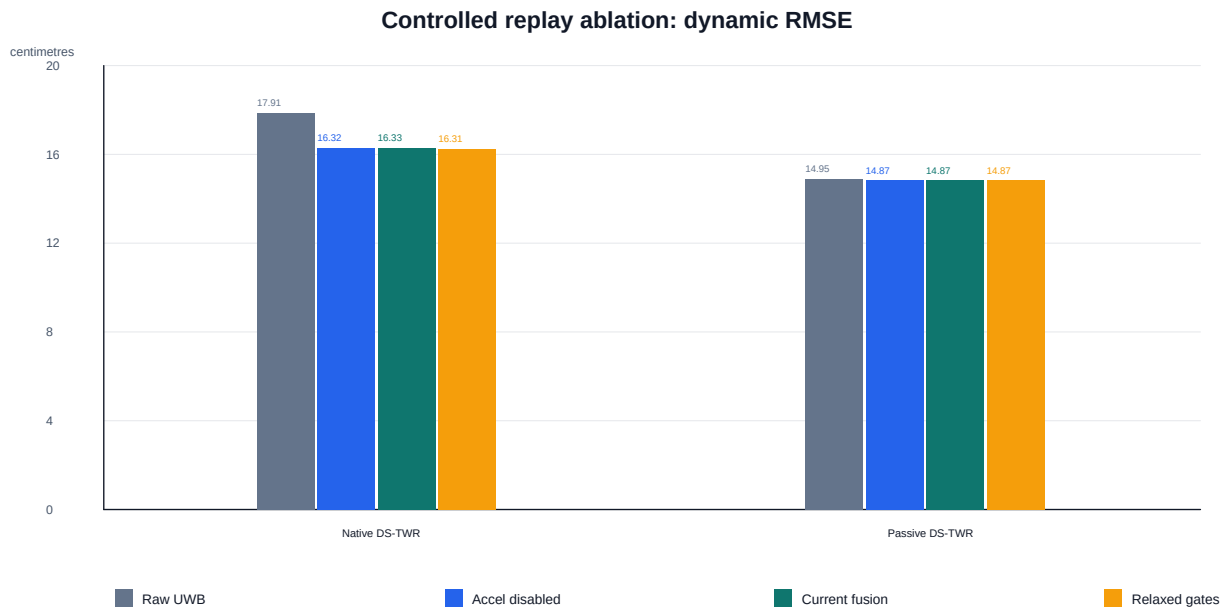


Figure 9: Dynamic RMSE under controlled replay ablation. The large raw-to- disabled change belongs to the common smoother; current acceleration is indistinguishable from disabled.

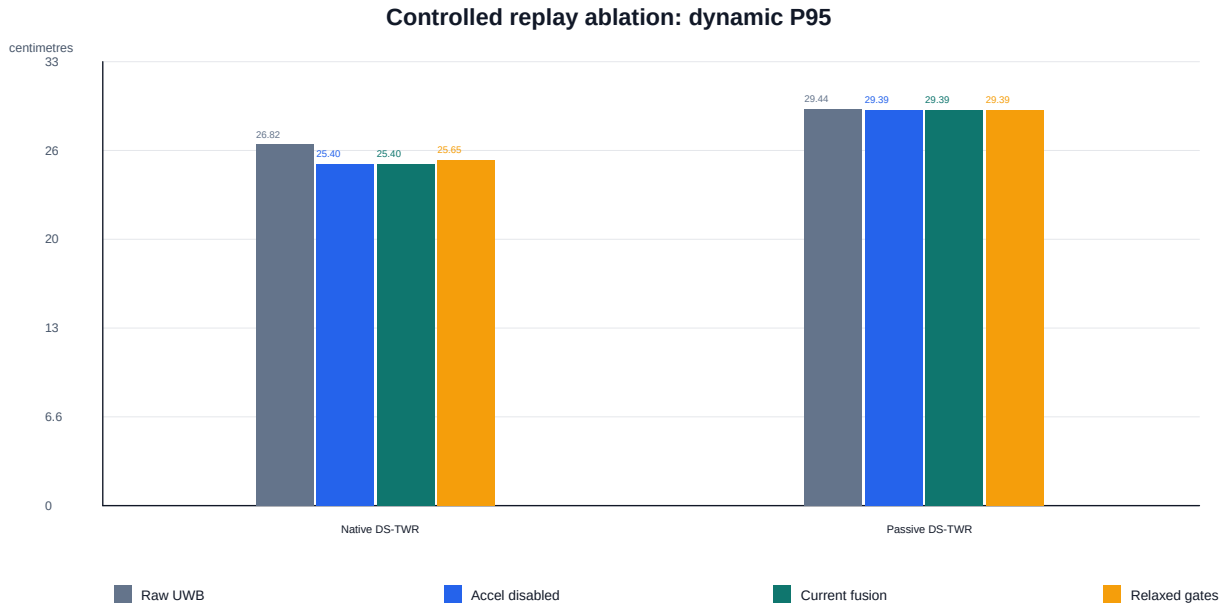


Figure 10: Dynamic P95 under the same ablation. Relaxing Native acceleration gates slightly worsens P95.

numerically identical even after 26903 samples are allowed to propagate. The current vector-acceleration path therefore fails the causal acceptance criterion.

9.3 Why acceleration was often unavailable

Protocol	propagated / valid	dominant exclusive gates
Native DS-TWR	20449 / 108563 = 18.84%	yaw stale 69.30%; yaw absent 9.28%
FlexTDOA	0 / 98805	yaw absent 39.20%; yaw stale 34.87%; bias missing 25.93%
Passive DS-TWR	0 / 78893	yaw stale 56.92%; yaw absent 25.53%; bias missing 17.55%

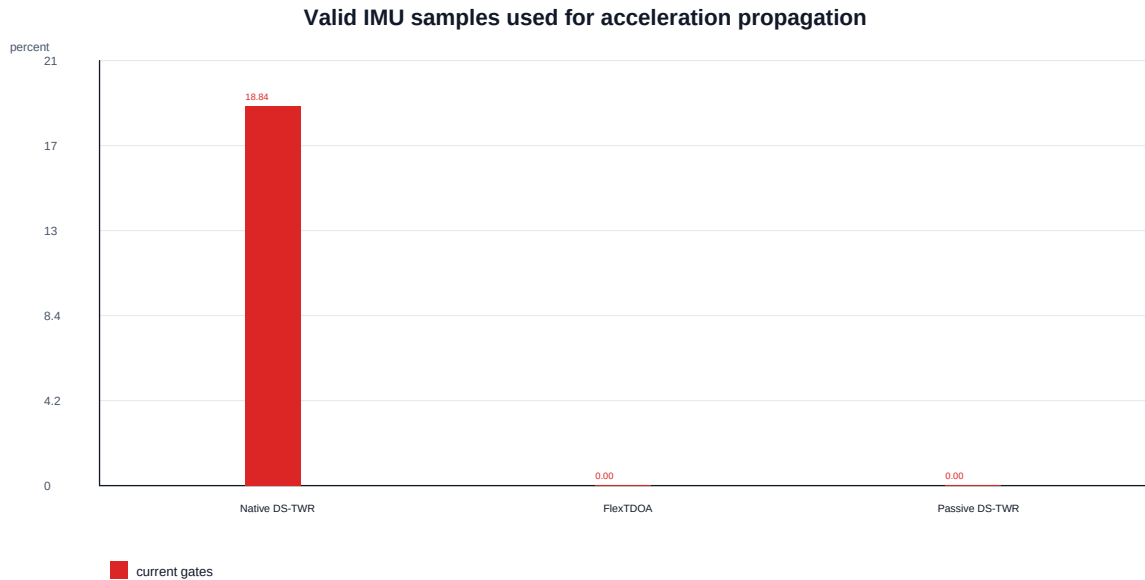


Figure 11: Current-gate acceleration propagation coverage. Zero does not mean that IMU transport failed; it means vector acceleration was blocked.

Flex has an additional replay epoch defect: an IMU timer backstep is translated into an artificial 2^{32} ms span and triggers a reset. Its zero-propagation figure cannot be attributed only to the normal gates. The current orientation mapping also assumes body +X follows travel, which is false for arbitrary handheld rotation and for the simulated throw.

10 Root-cause findings beyond the filter

10.1 Missing radio measurement timestamp

The current position wire format contains a 32-bit `uptime.ms` assigned when telemetry is submitted, not the radio measurement time. Dashboard and replay therefore fall back to receive time. The required correction is a versioned position stream carrying a 64-bit `measurement_monotonic.us`, derived from the midpoint of the radio RX host timestamps and preserved through the dashboard archive.

10.2 Scheduling risks on the UWB core

The UWB task runs on core 1 at priority 20 and installs the global GPIO ISR service. BNO085 starts later on core 0, but its 500 Hz handler can consequently execute on core 1 and perform a GPTimer read plus a critical section. Telemetry also runs on core 1; its dequeue path scans up to 1024 PSRAM ring entries under a critical section. Task priority cannot preempt code while interrupts are masked. Both paths can increase missed radio deadlines and should be A/B tested after restoring an O(1) FIFO or separate priority queues.

10.3 What the IMU can safely do now

Until a physical body-to-UWB mount calibration and a stable yaw source exist, the safe uses are stationary detection, bias observation, ZUPT, motion-state classification and adaptive process noise. Vector acceleration should remain in a shadow predictor and must not replace raw UWB positions.

11 Protocol assessment

Criterion	FlexTDOA	Native DS-TWR	Passive DS-TWR
Tag radio role	receive-only; scalable	active exchange with each anchor	receive-only
Dynamic independent rate	12.90 Hz	best: 20.86 Hz	13.33 Hz after overlap removal
Dynamic worst gap	2.55 s	best: 1.02 s	2.68 s
Static local RMS	2.13 cm	2.02 cm	best: 1.78 cm
Current fused replay	rejected: P95 worsens 2 mm	generic pass, but not an IMU gain	rejected on per-tag RMSE/P95
Operational role	primary scalable mode after timestamp fix	coherent diagnostic baseline	useful passive mode; continuity still matters

No protocol rewrite is justified by these data. The largest immediate gains are expected from correct measurement timing, reduced UWB-core interference and radio-loss diagnostics, not from more aggressive inertial integration.

12 Limitations

- The three dynamic routes are separate walks, not one synchronized path.
- RTK is an in-system comparison reference and uses software association.
- GNSS and UWB antenna lever arms are not surveyed.
- The original static blocks fail the common-reference consistency gate; only local precision is comparable across them.
- One stationary point does not cover GDOP, polygon edges, NLOS or tag orientation.
- The Passive dynamic container lacks `capture_end`; its data are included but its completeness status is not promoted.
- The causal ablation is available for Native and Passive. Flex requires a replay after the timer-epoch defect is corrected.

13 Recommended next validation sequence

1. Add and validate the 64-bit radio measurement timestamp end-to-end.
2. Replace telemetry's O(1024) critical-section scan with O(1) queues and A/B test IMU off, acquisition-only and full streaming.
3. Keep yaw-free IMU assistance limited to stationary detection, ZUPT and adaptive process noise.
4. Calibrate the physical IMU-to-UWB rotation before using vector acceleration; do not infer body +X from arbitrary walking direction.

5. Repeat one marked dynamic route for all protocols and collect at most 30 s static at the same point without restarting the RTK reference.
6. Accept fusion only when current beats acceleration-disabled on held-out RMSE and P95, with no larger maximum or dead-zone count.
7. After timing is trustworthy, investigate channel-9 antenna delay and per-link tag-anchor biases to approach the absolute ± 2 cm target.

14 Final conclusion

The campaign is a useful negative result. High-rate IMU transport works and stationary raw UWB remains centimetric, but the accelerometer has not improved dynamic positioning. The visually calmer fused trace is primarily a smoother. Calling the 17.91 to 16.33 cm Native change an accelerometer improvement would be incorrect: the acceleration-disabled replay already reaches 16.323 cm.

Raw UWB remains the production and audit reference. Fusion stays optional and experimental until measurement timing, orientation calibration and causal held-out tests demonstrate a real improvement.

15 Reproduction and audit artifacts

The report directory contains:

- `analysis.summary.json`, `capture.metrics.csv`, `replay.metrics.csv`, `rtk.pairs.csv` and alignment CSVs;
- `ablation.metrics.csv` and the script that generates the additional ablation figures;
- eleven SVG/PDF figures, this TeX source and the compiled PDF;
- source paths to the immutable JSONL captures. Raw archives were not duplicated into this directory.

The six matrix captures are in `reports/imu_fusion_final_validation_20260811` and `reports/imu_fusion_static_20260811`. The current Passive regression control is `passive.ds.static_raw_regression.01.passive.ds.jsonl`.

```
python tools/uwb_dynamic_static_report.py ... --include-incomplete
python reports/imu_fusion_ablation_20260811/generate_additional_figures.py
pdflatex -interaction=nonstopmode report.tex
```

No raw capture is modified during report generation.